Jacob Dowell

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Summary

Hands-on Full-Stack Engineering Lead with 7+ years at Charles Schwab designing and modernizing customer-facing financial platforms in highly regulated environments. I drive legacy modernization and cloud migration, shaping scalable API- and event-driven architectures for high availability, fault tolerance, and low latency. I leverage AI-assisted development and data-driven analysis to improve delivery speed and code quality while mentoring engineers and partnering with business and risk stakeholders to deliver compliant, resilient solutions.

Professional Experience

Lead Software Engineer at Charles Schwab **June 2018 - Present**

Software Engineer IV (Lead Back-End Engineer) — Nov 2024 – Present

- Technical lead modernizing a financial-derivatives risk-monitoring platform, migrating 17 legacy .NET Framework console applications into scalable .NET (C#) web services to support time-sensitive risk workflows.
- Own end-to-end architecture for services and data flows, designing and authoring Software Architecture Documents (SADs) and presenting to Local Governance to align with enterprise standards and risk controls.
- Built a reusable Kafka wrapper library that standardizes configuration, error handling, and observability, enabling 17+ applications to adopt consistent event-driven patterns and reducing boilerplate and defects.
- Led migration of 17 legacy applications to GitHub, defining CI workflows with GitHub Actions for automated build, test, and deployment, improving release reliability, traceability, and developer productivity.
- Introduced AI-assisted development and data-driven reviews into the team's workflow, accelerating delivery while maintaining code quality, architectural consistency, and alignment with best practices.

Software Engineer III (Lead Front-End Engineer) — Sep 2021 – Nov 2024

- Technical lead for Angular single-page applications (SPAs) on Schwab's Rep Technology platform used by 20,000+ call center reps, delivering high-availability, low-latency user experiences for time-sensitive client servicing workflows.
- Led front-end delivery for major integration programs (USAA, TDA, Account Number Expansion), defining end-to-end flows, authoring SADs, and presenting designs to Local Governance to align with enterprise standards and control requirements.
- Refactored middle-tier service layers into RESTful APIs, simplifying contracts between the UI and back-end systems and improving maintainability and performance.
- Mentored junior developers on front-end architecture, coding practices, and production support, while establishing coding standards and review practices that raised quality and consistency across the team.
- Completed Schwab's Aspiring Leaders Program, further developing coaching, communication, and delivery leadership skills.

Software Engineer II (Front-End .NET/Angular Engineer) — Mar 2020 – Sep 2021

- Modernized a legacy WinForms application into a cloud-hosted Single Page Application, improving usability for call center reps, reducing friction in daily workflows, and making the platform easier to evolve and deploy.
- Partnered with UX and product teams to redesign key service journeys, enabling faster, more intuitive client interactions and reducing manual steps for reps.
- Implemented CI/CD pipelines using Bitbucket, Bamboo, and Pivotal Cloud Foundry, automating builds, tests, and deployments to reduce manual effort and release risk.
- Introduced and expanded unit testing with MSTest and Jasmine, contributing to 90% code coverage and more reliable releases.

- Worked closely with operations and support teams to instrument logging (Splunk) and monitoring, enabling proactive detection and troubleshooting of production issues.

Software Engineer I (Back-End Java/Spring Engineer) — Jun 2018 – Mar 2020

- Refactored a high-traffic legacy API into modular, maintainable services, improving response times and easing future enhancements.
- Built a Redis-based microservice to cache frequently accessed data, reducing database load and improving latency for downstream applications.
- Graduated from Schwab's NERD program, gaining structured experience in enterprise SDLC, code quality, and collaboration practices.

Programming Fundamentals Tutor at Texas State University **Feb 2017 – May 2018**

- Tutored undergraduate students in programming fundamentals (C# and Java), helping them break down problems and build confidence in their coding skills.
- Reviewed, debugged, and refactored student code, explaining core concepts like control flow, data structures, and object-oriented design in accessible terms.
- Collaborated with faculty to understand course expectations and tailor tutoring sessions to reinforce current assignments and exam topics.

Bridal Consultant at Bed Bath & Beyond **July 2014 – August 2016**

- Trained 10 new employees on corporate policies, customer workflows, and point-of-sale processes.
- Created bridal registries according to budgetary goals to maximize revenue
- Generated up to 10% of the store's daily income through instore online transaction orders

CBRN Reconnaissance Team Leader at United States Army **September 2009 – March 2013**

- Managed a reconnaissance team of three to ensure mission readiness and individual improvement
- Assisted commanding officers in planning missions according to company procedures, minimizing risk

Skills

- **Languages**: C#, HTML, Java, JavaScript, CSS/SASS, TypeScript
- **Frameworks**: .NET, .NET Framework, Angular 2+, Node.js, Spring
- **Libraries**: jQuery, Lodash, NgRx, React, RxJS
- **Cloud**: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), Pivotal Cloud Foundry (PCF)
- **Databases**: Aerospike, MongoDB, Redis, SQL Server
- **Testing**: Jasmine, Jest, JUnit, Karma, MSTest, xUnit
- **Tools & Technologies**: Agile, Bamboo, Bitbucket, CI/CD, GitHub, Kafka, Postman, RabbitMQ, Scrum, Splunk

Certifications

- Google Associate Cloud Engineer — March 2025
- Salesforce JavaScript Developer I — May 2024
- Microsoft AZ-900 Azure Fundamentals — January 2022
- Microsoft 98-361 Software Development Fundamentals (C#) — September 2020
- Professional Scrum Master I (PSM I) — February 2019
- Microsoft 98-388 Introduction to Programming (Java) — September 2018

Education

McCoy College of Business: Texas State University, San Marcos, TX **August 2014 – May 2018**

- Bachelor of Business Administration in Computer Information Systems & Quantitative Methods
- Cumulative GPA: 3.54/4.00
- Marvin A. Johnston Excellence Award
- Outstanding Undergraduate Student Award in CIS